CRYSTAL
PRACTICE MANAGEMENT

CRYSTAL PM

HOW-TO GUIDE

Set up a Crystal Meet door sign panel

An SMT1019 wall panel by the door shows the room's door sign full screen and lights the bar round its edge green, yellow or red with the room's status.

Do this for each panel, after the room itself is set up (the guide "Set up a Crystal Meet room"). Plan about 15 minutes per panel. You install one small app, the Crystal Meet Door Sign app, from a computer over a USB-C cable; after that the panel needs only power and the network. The door sign itself is the same page any screen can show at `http://<device address>:8080/sign` : the app adds the light bar, starts with the panel and keeps people in the hallway out of Android.

STEP 1
Get the app
Build it once from the repository.

STEP 2
USB debugging
Developer options on the panel, then the cable.

STEP 3
Install
One adb command.

STEP 4
Make it the panel's app
It starts with the panel; Home comes back to it.

STEP 5
First setup
The room device's address and a PIN.

STEP 6
Check the lights
Green, yellow, red, off.

BEFORE YOU BEGIN

The panel. An SMT1019 10.1-inch wall panel (Android 14, made by Apolosign/Hopestar), powered by PoE from a switch or injector, or by its own power supply, and on the office network by Ethernet or Wi-Fi. It must reach the room device on port 8080. Another Android tablet runs the app as well, without the light bar.

A computer with Android Platform Tools (adb) and a USB-C data cable. On Windows, in PowerShell: `winget install Google.PlatformTools` , then open a new PowerShell. On Linux or WSL: `sudo apt install adb` (WSL reaches the panel over the network only, not over USB).

The room device's address, for example `10.0.0.21` : the fixed address the room guide reserved for it.

1 Get the app

- **Build it once.** The app's code is in `android/door-sign` in the Crystal Meet repository. On a computer with JDK 17 and the Android SDK (platform 34, build-tools 34.0.0, NDK 26.3.11579264, CMake 3.22.1), type `cd android/door-sign`, then `echo "sdk.dir=$ANDROID_HOME" > local.properties`, then `./gradlew testDebugUnitTest assembleDebug`.
- **The file to install** is `android/door-sign/app/build/outputs/apk/debug/app-debug.apk`. Copy it to the computer you install from and call it `door-sign.apk`; the commands below use that name.
- **Build every panel's app on the same computer.** Android accepts an update only from the build that installed the app first. An app built elsewhere has to replace the old one with `adb uninstall to.croom.doorsign` first, which also clears the panel's address and PIN.

You should now have one APK file of about 45 KB.

2 Turn on USB debugging

- **Developer options.** On the panel open **Settings › About tablet** and tap **Build number** seven times until it says you are a developer. Then **Settings › System › Developer options**, and switch on **USB debugging**. On a new SMT1019 it is often on already.
- **If the panel is locked in another app** (its kiosk mode was switched on in Android's settings), hold **Back** and **Recents** together for two seconds, or swipe up from the bottom and hold, to leave it first.
- **Connect the cable** from the computer to the panel's USB-C port. The panel asks **Allow USB debugging?**: tick **Always allow from this computer** and tap **Allow**.
- **Check.** On the computer type `adb devices`. The panel is listed with the word `device` after its serial number.

You should now see one line ending in device. "unauthorized" means the prompt on the panel is still waiting.

3 Install the app

- In the folder with the file, type `adb install -r door-sign.apk`. It prints `Success`. The same command installs a newer build later and keeps the panel's address and PIN.

4 Make it the panel's app

- **The home app.** Type `adb shell cmd package set-home-activity to.croom.doorsign/.MainActivity`. The panel then opens the door sign when it starts, and its Home button comes back to it.
- **The start-up app.** Type `adb shell setprop persist.sys.bootapp to.croom.doorsign/to.croom.doorsign.MainActivity`. The SMT1019 opens this app at start-up in place of whatever was set there before (Chrome, on some panels).
- **Open it now** with `adb shell am start -n to.croom.doorsign/.MainActivity`, or press Home on the panel.

You should now see the setup screen on the panel, headed Door sign setup.

5 First setup on the panel

- **Crystal Meet server.** Type the room device's address and port, for example `10.0.0.21:8080`. The app adds `http://` and `/sign`; a full address such as `http://10.0.0.21:8080/sign` is kept as it is.
- **PIN.** Choose 4 to 8 digits and keep them with the room's other passwords. The PIN is what lets anyone leave the sign later.
- Tap **Save**. The door sign fills the screen, with no Android bars.

You should now see the room's name and status, and the light bar in the status colour within a few seconds.

6 Check the lights

The bar round the panel follows the status the sign shows:

Green	Yellow	Red	Off
Free.	A booking starts within 10 minutes.	In a meeting, or booked.	The sign cannot reach the room device, or is still loading.

- **Test the bar itself.** Hold the top-left corner of the sign for 5 seconds, enter the PIN and use **Green**, **Yellow**, **Red** and **Off** under **Test the light bar**. Press Back to return to the sign, which puts the room's colour back.
- **Test with a booking.** Book the room for a few minutes from now: the bar turns yellow, then red when the booking starts, then green again when it ends. The sign checks the room every 5 seconds and the bookings every minute.

You should now see the bar change with the sign's own colour, and never disagree with it.

AFTER SETUP

Everyday use

- **Open setup.** Hold the top-left corner of the sign, over the Crystal logo, for 5 seconds, and enter the PIN. A shorter touch does nothing, so the sign's own **This week** button stays the only thing to press in the hallway.
- **Change the address or the PIN** in setup and tap **Save**. Leave the new PIN empty to keep the old one.
- **Leave the sign.** In setup tap **Exit to Android**: the panel's own home screen opens with its status bar, and the light bar goes off. Press Home to bring the sign back.
- **After a power cut** the panel starts straight into the sign. Until the room device answers, the sign says **Can't reach Crystal Meet** with the address, the lights stay off, and it tries again every 10 seconds.
- **Background and football scores** are chosen on the table screen under **Options**, then **Door sign**, as for any door sign (the room guide's section The door sign).
- **Update the app** with `adb install -r door-sign.apk` from the computer that built the first one; the address and PIN stay.

TAKING A PANEL BACK

Return the panel to a normal tablet

- Over the cable type, in this order: `adb shell cmd package set-home-activity com.android.launcher3/.uioverrides.QuickstepLauncher`, then `adb shell setprop persist.sys.bootapp ""`, then `adb shell am broadcast -a elc.view.show` (the status bar), then `adb uninstall to.croom.doorsign`.

IF SOMETHING IS OFF

Troubleshooting

"Can't reach Crystal Meet at …"

The panel cannot open the address. Open the same address in a browser on a computer on the panel's network. No answer there: the room device is off, the address or port is wrong, or the panel is on a network (a guest Wi-Fi, another VLAN) that cannot reach the room device. Fix the address in setup.

The sign shows, the lights stay off

"Sign not connected" on the sign turns the lights off on purpose. If the setup screen's test buttons do nothing either, the panel is not an SMT1019 or its light controller is missing; `adb logcat -s DoorSign` says which, and shows every change (`Light bar: free`).

No internet over the PoE cable

Ethernet is on in Android already, so the fault is the cable or what feeds it. `adb shell ip link show eth0` says `NO-CARRIER` when no data link is there. Check that the PoE injector's data-in port is cabled to the switch, or that the switch port is up and its PoE on; try another cable (all 8 wires) and another port. Wi-Fi works meanwhile.

The status bar shows at the top

The app hides the SMT1019's status bar each time the sign comes up. If it is still there, press Home, or type `adb shell am broadcast -a elc.view.hide` .

Forgotten PIN

Over the cable type `adb shell pm clear to.croom.doorsign` . The app forgets its address and PIN and opens the first setup again.

adb install says UPDATE_INCOMPATIBLE

The app on the panel came from another build computer. Type `adb uninstall to.croom.doorsign` , install again, and redo steps 4 and 5.

GOOD TO KNOW

The panel talks only to the room device, over plain HTTP on the office network. The page tells the app the room's status by name (free, soon, occupied, offline) and the app alone chooses the colours, so a door sign in a browser elsewhere is unchanged. The light bar is driven the way the panel's own LED test drives it, through `/dev/ledjni` .